#14Days14Chapters

DAY 10

1 Feb

in-depth.

14 Feb

15 Feb

16 Feb

Maha Marathons.

17 Feb → The legendary 4AM Live!

[Aim: 100/100 in Maths]

POLYNOMIALS

What is a Polynomial?

$W = \{0, 1, 2, 3 - - -\}$

$\underbrace{\text{Polynomial}}$ → Type of A.E. → $\boxed{[\text{variable}]^{\text{whole no.}}}$

Identify which is a polynomial?

(i) $x^{2} - 2$ ✓

(ii) $\frac{1}{x} - 4 \Rightarrow x^{-1} - 4$ wα ✗

(iii) $\sqrt{x} - 4 \Rightarrow x^{\frac{1}{2}} - 4$ ✗

(iv) $2\sqrt{2}x^{1} - 3$ ✓

(v) $2x^{2} - 3x^{1} + 4y^{1} - 3z^{1} + \sqrt{8}$ ✓

(vi) $(x-1)^2 \Rightarrow x^{2} + 1 - 2x^{1}$ ✓

(vii) $\frac{\cancel{x^2}}{\cancel{x^3}} \Rightarrow \frac{1}{x} \Rightarrow x^{-1}$ ✗

#LP: $P(x) = \underbrace{5x^3} - \underbrace{4x^2} + \underbrace{7x} - \underbrace{9}$

Is it a polynomial? If yes, then answer the following:- Yes

i. No. of Terms= 4 terms

ii. Coefficient of x^3 = 5

iii. Coefficient of x^2 = -4

iv. Coefficient of x = 7

v. Constant term= -9

vi. Degree of Polynomial= 3

Variable की highest power

Types of Polynomials:

(i) on basis of no. of terms:-

- $n = 1 \Rightarrow$ monomial (eg:- x, 2, x^2, x^4)
- $n = 2 \Rightarrow$ binomial (eg:- $x+2$, x^2-4, x^3+x)
- $n = 3 \Rightarrow$ trinomial (eg:- x^2+x-3, x^2+x+1)

On basis of degree:

- $D=1 \Rightarrow$ Linear (eg: x^1+1, x^1, $x-\sqrt{2}$)
- $D=2 \Rightarrow$ $\boxed{\text{Quadratic}}$ (eg: x^2-4x+3, x^2, $x^2-\sqrt{2}$)
- $D=3 \Rightarrow$ cubic (eg: x^3-x, x^3-x^2+4, $3x^3+4x^2-3x+2$)

$$\boxed{2x^3 - 3x^2}$$

no. of terms ⇒ 2 → binomial

Degree ⇒ 3 → cubic

→ cubic binomial

$$\boxed{x^2 - 2x + 3}$$

n → 3 → trinomial

D → 2 ⇒ Quadratic

⇒ Quadric trinomial

Zero polynomial :- 0

* degree of zero polynomial ⇒ $0x^{0,1,2 \ldots}$

$\boxed{\text{Not defined}}$

Constant Polynomial :- 2 , 3 , π . . .

* degree of constant polynomial → $2x^0$

$\boxed{D = 0}$

#LP: Assertion (A): Degree of zero polynomial is not defined. T

Reason (R): Degree of a non-zero constant polynomial is 0. T

(CBSE 2024)

(b) ✓

Zeroes / roots / solution of Polynomial:

↳ variable की वो value जिससे $\boxed{Poly = 0}$ हो जाए !

eg: $P(x) = x - 4$

put $x = 4$

$P(4) = 4 - 4$

$= 0$

$\therefore$ 4 is zero of $P(x)$.

eg: $g(x) = \frac{x}{4} - \frac{3}{8}$

↳ $\boxed{g(x) = 0}$

$\frac{x}{4} - \frac{3}{8} = 0$

$\frac{x}{\cancel{4}} = \frac{3}{\cancel{8}_2}$

zero of $g(x)$ → $x = 3/2$

Next Toppers

#LP: If one of the zeroes of the quadratic polynomial $(a-1)x^2+ax+1$ is -3, then the value of a is: (2024)

(a) $-\frac{2}{3}$ (b) $\frac{2}{3}$ (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

let $P(x) = (a-1)x^2+ax+1$

$$P(-3) = 0$$

$$(a-1)(-3)^2 + a(-3)+1 = 0$$

$$9(a-1) - 3a+1 = 0$$

$$9a - 9 - 3a+1 = 0$$

$$6a - 8 = 0$$

$$\cancel{6}^{3}a = \cancel{8}^{4}$$

$$\boxed{a = \frac{4}{3}}$$

#LP: . If one of the zeroes of a quadratic polynomial $(k - 1)x^2 + kx + 1$ is -3, then the value of k is (2022)

(a) 4/3

(b) -4/3

(c) 2/3

(d) -2/3

let $p(x) = (k-1)x^2 + kx + 1$

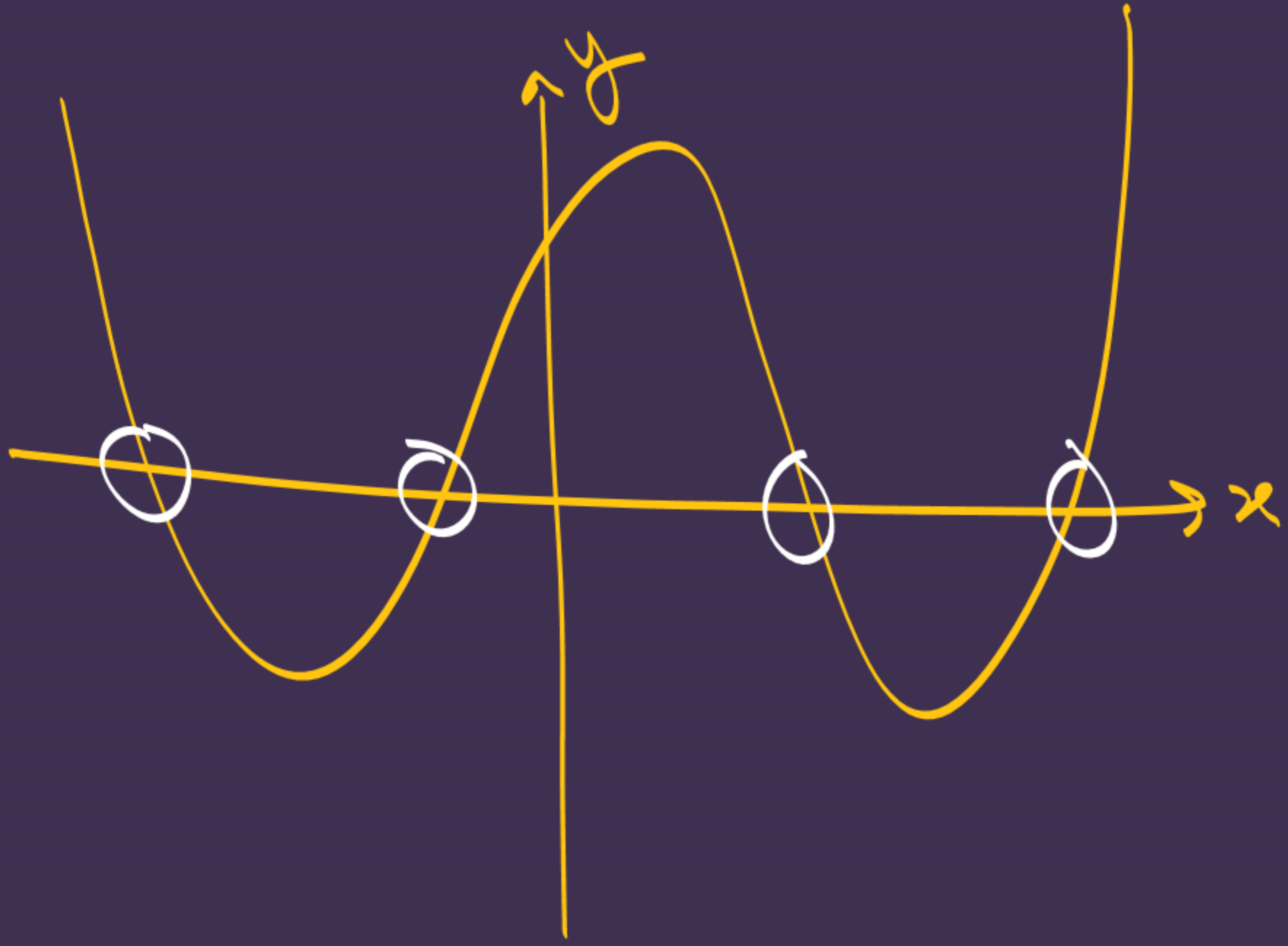
$$p(-3) = 0$$

$$(k-1)(-3)^2 + k(-3) + 1 = 0$$

$$\boxed{k = \frac{4}{3}}$$

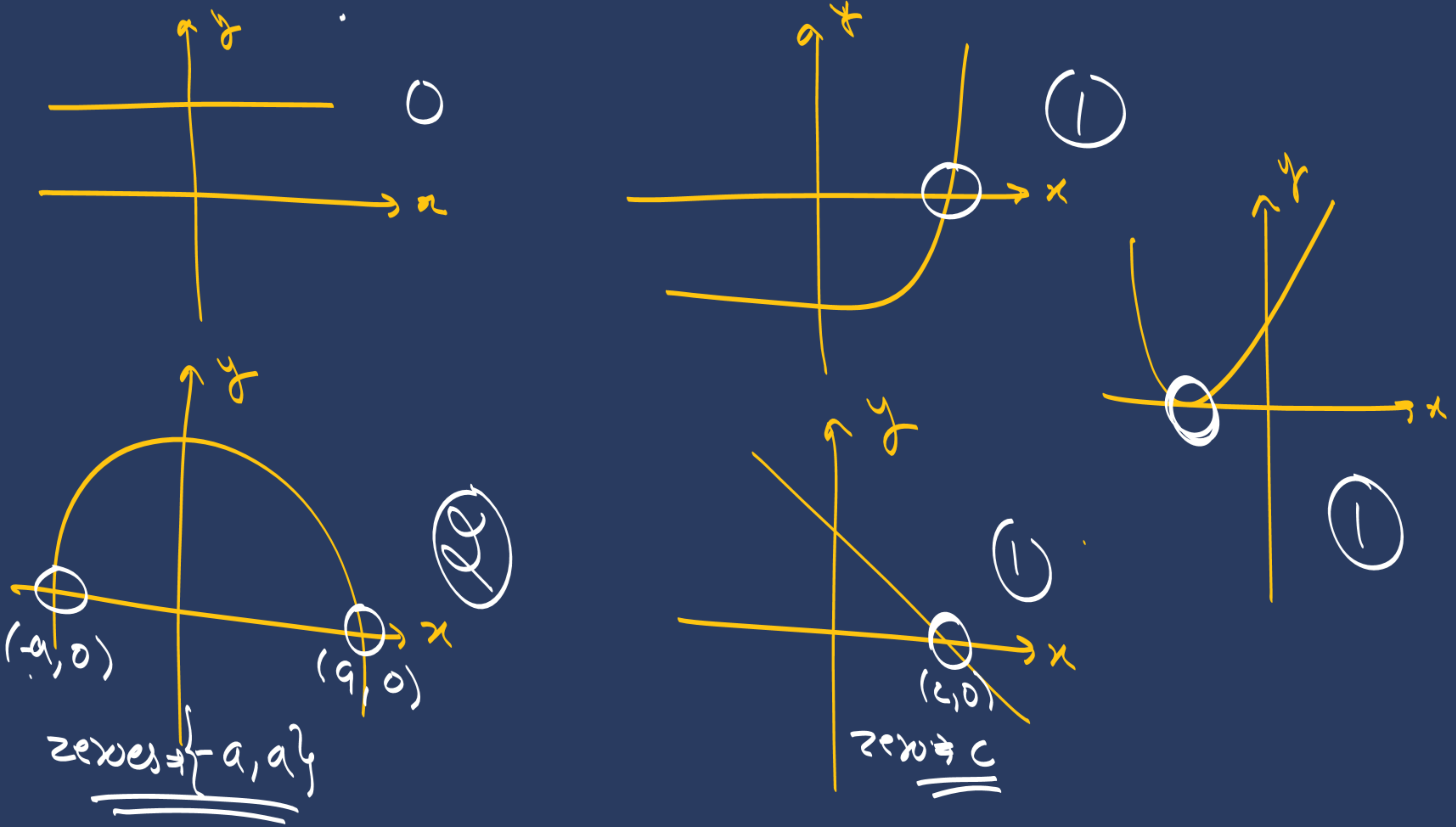
No. of Zeroes of Polynomials by Graph:

Polynomial ka graph jitni baar x-axis ko touch/ intersect karega utne no. of zeroes honge



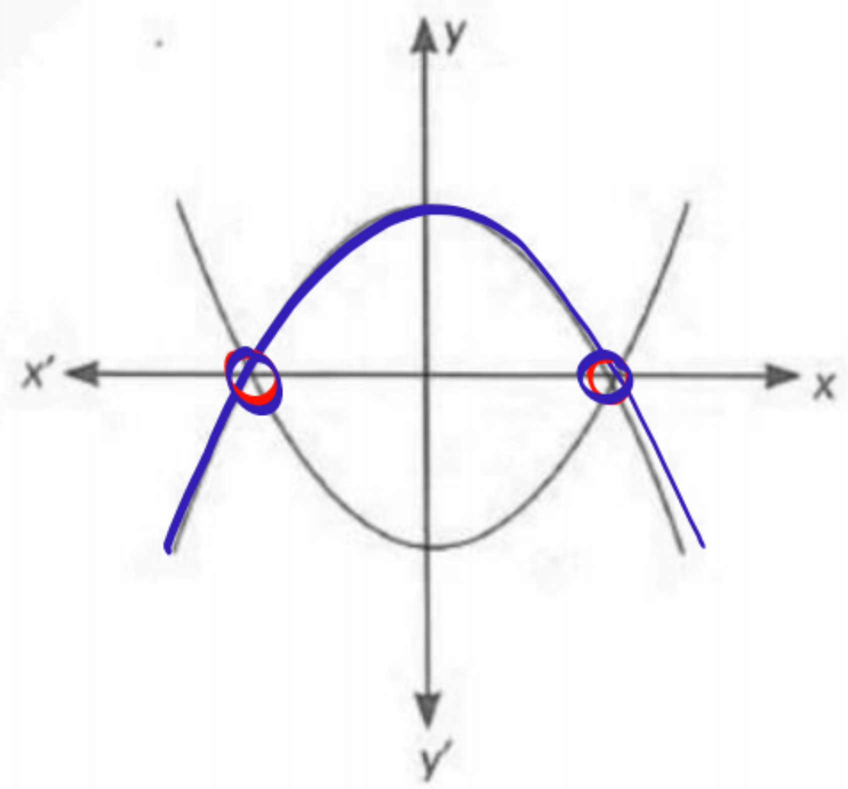
No of zeroes

$\Rightarrow 4$



y
x
0
y
x
1
y
x
1
y
x
2
(-a, 0)
(a, 0)
zeroes = {-a, a}
y
x
1
(c, 0)
zero = c

#LP: Two polynomials are shown in the graph below. The number of distinct zeroes of both the polynomials is: (2025)



(a) 3

(b) 5

(c) 2

(d) 4

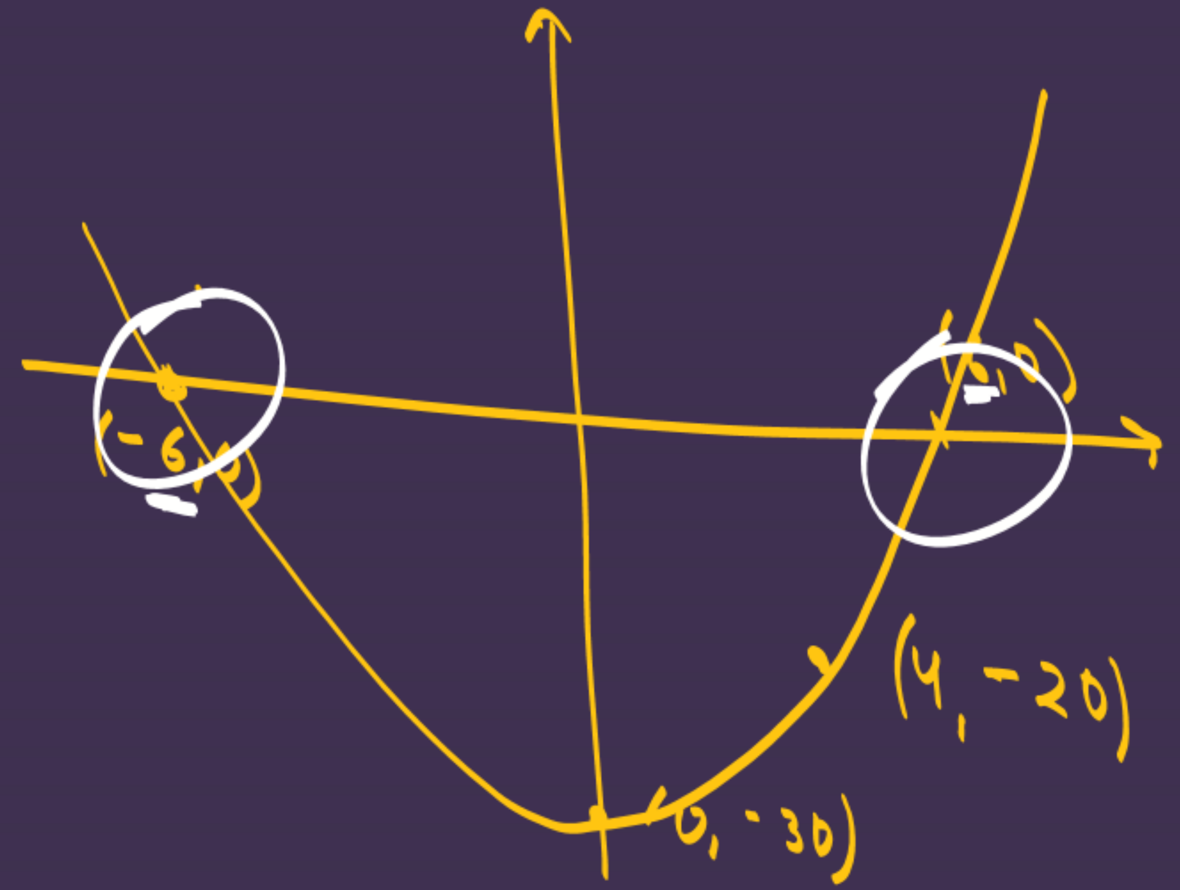
#LP: The graph of a quadratic polynomial $p(x)$ passes through the points $(-6, 0)$, $(0, -30)$, $(4, -20)$ and $(6, 0)$. The zeroes of the polynomial are

A) $-6, 0$

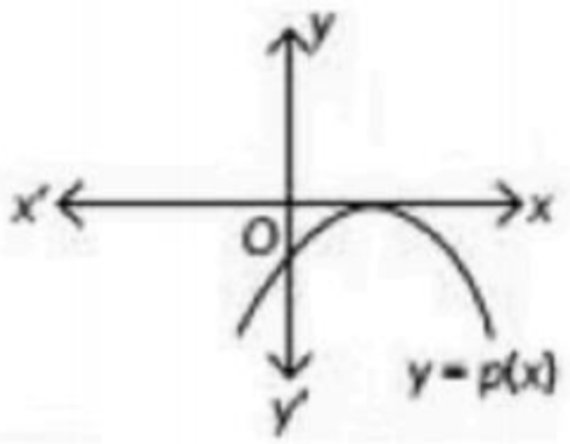
B) 4,6

C) $-30, -20$

D) $-6, 6$



#LP: The graph of y = p(x) is given, for a polynomial p(x). The number of zeroes of p(x) from the graph is (2023)



(a) 3

(b) 1

(c) 2

(d) 0

Middle Term Splitting Method to find Zero of a Quadratic Polynomial :

$$3x^2 - 11x - 4 = 0$$

$$3x^2 - 12x + 1x - 4 = 0$$

$$3x(x-4) + 1(x-4) = 0$$

$$x-4\{3x+1\} = 0$$

$$(x-4)(3x+1) = 0$$

$x - 4 = 0$ | $3x + 1 = 0$

$\boxed{x = 4}$ | $\boxed{x = -\frac{1}{3}}$

$$\begin{array}{r|l} 2 & 12 \\ 2 & 6 \\ 3 & 3 \\ & 1 \end{array}$$

4, 3

$$p + q = -11$$

$$pq \Rightarrow -12$$

$$x^2 + 10x + 24 = 0$$

$$x^2 + 6x + 4x + 24 = 0$$

$$x(x+6) + 4(x+6) = 0$$

$$(x+6)(x+4) = 0$$

$$(x+6)(x+4) = 0$$

$x + 6 = 0$ | $x + 4 = 0$

$\boxed{x = -6}$ | $\boxed{x = -4}$ ✓

$$\begin{array}{r|l} 2 & 24 \\ 2 & 12 \\ 2 & 6 \\ 3 & 3 \\ & 1 \end{array}$$

4, 6

$$p + q = 10$$

$$pq = 24$$

#LP: **Zeroes of the polynomial** $\boxed{p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3}}$ are : **(2025)**

(a) $-\frac{2}{3}, -\frac{1}{7}$

(b) $-\frac{2}{7}, -\frac{1}{3}$

(c) $\frac{2}{3}, \frac{1}{7}$

(d) $\frac{2}{3}, -\frac{1}{7}$ ✓

$p(y) = 0$

$7y^2 - \frac{11}{3}y - \frac{2}{3} = 0$

$\Rightarrow \frac{21y^2 - 11y - 2}{3} = 0$

$\Rightarrow \boxed{21y^2 - 11y - 2 = 0}$

$p + q = -11$

$pq = -42$

$$\begin{array}{r|l} 2 & 42 \\ \hline 3 & 21 \\ \hline 7 & 7 \\ \hline & 1 \end{array}$$

14

$21y^2 - 14y + 3y - 2 = 0$

$7y(3y - 2) + 1(3y - 2) = 0$

$(3y - 2)(7y + 1) = 0$

$3y - 2 = 0$ | $7y + 1 = 0$

$y = \frac{2}{3}$ | $y = -\frac{1}{7}$

Relationship b/w zeroes and coefficients of a Quadratic Polynomial :

V.IMP

$$\boxed{ax^2+bx+c} \begin{cases} \alpha \\ \beta \end{cases}$$

Sum of roots $\rightarrow \alpha+\beta = -\frac{b}{a} = \frac{-(\text{coeff of } x)}{\text{coeff of } x^2}$

Product of roots $\rightarrow \alpha\cdot\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coeff of } x^2}$

Q: $3x^2 - 2x + 3 < \begin{matrix}\alpha \\ \beta\end{matrix}$

SOR $= \frac{-b}{a} = \frac{-(-2)}{3} \rightarrow \frac{2}{3} = \alpha + \beta$

POR $= \alpha \cdot \beta = \frac{c}{a} = \frac{\cancel{3}}{\cancel{3}}\ 1$

$\alpha \cdot \beta = 1$

Q: $-x^2 + 3x - 2 < \begin{matrix}\alpha \\ \beta\end{matrix}$

SOR $= \frac{-(3)}{-1} \rightarrow 3 = \alpha + \beta$

POR $= \alpha \cdot \beta = \frac{c}{a} = \frac{+2}{+1}$

$\alpha \cdot \beta = 2$

#LP: Find the zeroes of each of the following quadratic polynomial and verify the relationship between the zeroes and their coefficients:

(i) $g(s) = 4s^2 - 4s + 1$

$g(s) = 0$

$4s^2 - 4s + 1 = 0$

$4s^2 - 2s - 2s + 1 = 0$

$2s(2s-1) - 1(2s-1) = 0$

$(2s-1)(2s-1) = 0$

$2s - 1 = 0 \quad | \quad 2s - 1 = 0$

$s = \frac{1}{2} - \alpha \quad | \quad s = \frac{1}{2} - \beta$

SOR

$\alpha + \beta = \frac{-b}{a}$

LHS $\frac{1}{2} + \frac{1}{2} \Rightarrow 1$

RHS $\frac{-(-4)}{4} \Rightarrow 1$

LHS = RHS HP

$\alpha \cdot \beta = \frac{c}{a}$

LHS $\Rightarrow \frac{1}{2} \times \frac{1}{2} \Rightarrow \frac{1}{4}$

RHS $\Rightarrow \frac{1}{4}$

LHS = RHS HP

(ii) $g(x) = 6x^2 - 3 - 7x$

$g(x) = 0$

$6x^2 - 3 - 7x = 0$

$\boxed{6x^2 - 7x - 3 = 0}$

$6x^2 - 9x + 2x - 3 = 0$

$3x(2x-3) + 1(2x-3) = 0$

$(2x-3)(3x+1) = 0$

$2x - 3 = 0 \quad | \quad 3x + 1 = 0$

$x = 3/2 - \alpha \quad | \quad x = -1/3 - \beta$

Verification

SOR $\alpha + \beta = \frac{-b}{a}$

LHS $\frac{3}{2} + \left(-\frac{1}{3}\right) \Rightarrow \frac{9-2}{6} \Rightarrow \frac{7}{6}$

RHS $\frac{-(-7)}{6} \Rightarrow \frac{7}{6}$

LHS = RHS HP

POR $\alpha \cdot \beta = \frac{c}{a}$

LHS $\frac{3}{2} \times -\frac{1}{3} \Rightarrow -\frac{1}{2}$

RHS $\Rightarrow \frac{c}{a} \Rightarrow \frac{-3}{6} \Rightarrow -\frac{1}{2}$

LHS = RHS HP

(iii) $q(x) = \sqrt{3}x^2 + 10x + 7\sqrt{3}$

$q(x) = 0$

$\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$

$\sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = 0$

$\Rightarrow \sqrt{3}x^2 + \sqrt{3}\sqrt{3}x + 7x + 7\sqrt{3} = 0$

$\Rightarrow \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$

$(x + \sqrt{3})[\sqrt{3}x + 7] = 0$

$x + \sqrt{3} = 0$ | $\sqrt{3}x + 7 = 0$

$x = -\sqrt{3}$ | $\boxed{x = -\frac{7}{\sqrt{3}}}$

Verify

HW

(iv) $f(v) = v^2 + 4\sqrt{3}v - 15$

$f(v) = 0$

$v^2 + 4\sqrt{3}v - 15 = 0$

$p + q = 4\sqrt{3}$

$p \cdot q = -15$

$v^2 + 5\sqrt{3}v - \sqrt{3}v - 15 = 0$

$v(v + 5\sqrt{3}) - \sqrt{3}(v + 5\sqrt{3}) = 0$

$(v + 5\sqrt{3})(v - \sqrt{3}) = 0$

$v + 5\sqrt{3} = 0$ | $v - \sqrt{3} = 0$

$\boxed{v = -5\sqrt{3}}$ | $\boxed{v = \sqrt{3}}$

#K³B

$$x^2+5x+6=0$$ ← 3 terms

$$\underbrace{x^2+2x}+\underbrace{3x+6}=0$$ ← 4 terms

$$x(x+2)+3(x+2)=0$$ ← common करते है

$$(x+2)(x+3)=0$$

(V) $f(x) = x^2 - (\sqrt{3}+1)x + \sqrt{3}$

$$f(x) = 0$$

$$x^2 - (\sqrt{3}+1)x + \sqrt{3} = 0$$

$$\underbrace{x^2 - \sqrt{3}x} - \underbrace{x + \sqrt{3}} = 0 \leftarrow$$ 4 terms (already middle term split).

$$x(x-\sqrt{3}) - 1(x-\sqrt{3}) = 0$$

$$(x-\sqrt{3})[x-1] = 0$$

$x - \sqrt{3} = 0$ | $x - 1 = 0$

$x = \sqrt{3}$ | $x = 1$

#LP: Find the zeroes of the quadratic polynomial $\boxed{2x^2 - (1 + 2\sqrt{2})x + \sqrt{2}}$ and verify the relationship between the zeroes andcoefficients of the polynomial.

[CBSE 2025]

$$2x^2 - (1 + 2\sqrt{2})x + \sqrt{2} = 0$$

$$\underbrace{2x^2 - x} - \underbrace{2\sqrt{2}x + \sqrt{2}} = 0 \leftarrow \text{4 terms}$$

$$x(2x - 1) - \sqrt{2}(2x - 1) = 0$$

$$(2x - 1)[x - \sqrt{2}] = 0$$

$2x - 1 = 0$ | $x - \sqrt{2} = 0$

$x = \frac{1}{2}$ | $x = \sqrt{2}$

#LP: For what value of k, the product of zeroes of the polynomial $kx^2 - 4x - 7$ is 2? (2024)

(a) $-\frac{1}{14}$ (b) $-\frac{7}{2}$ (c) $\frac{7}{2}$ (d) $-\frac{2}{7}$

$$\boxed{kx^2 - 4x - 7}$$

$$POR = 2$$

$$\frac{c}{a} = 2$$

$$\frac{-7}{k} = 2 \Rightarrow k = -7/2$$

#LP: If α and β are the zeroes of the polynomial $p(x) = \boxed{x^2 - ax - b}$, then the value of $(\alpha + \beta + \alpha\beta)$ is equal to: (2025)

(a) $a + b$

(b) $a - b$

(c) $-a - b$

(d) $-a + b$

$$\boxed{x^2 - ax - b} \quad \alpha, \beta$$

$$\alpha + \beta = -\frac{(-a)}{1}$$

$$\boxed{\alpha + \beta = a}$$

$$\alpha \cdot \beta = -\frac{b}{1}$$

$$\boxed{\alpha \cdot \beta = -b}$$

$$\underline{\alpha + \beta} + \alpha\beta$$

$$a + (-b)$$

$$\boxed{a - b}$$

#LP: If α, β are the zeroes of a polynomial $p(x) = \boxed{4x^2 - 3x - 7}$, then $(1/\alpha + 1/\beta)$ is equal to (2023)

(a) 7/3

(b) -7/3

(c) 3/7

(d) -3/7 ✓

$\boxed{4x^2 - 3x - 7} < \begin{matrix}\alpha \\ \beta\end{matrix}$

$\alpha + \beta = -\frac{(-3)}{4}$

$\boxed{\alpha \cdot \beta = -\frac{7}{4}}$

$\alpha + \beta = 3/4$

$\frac{1}{\alpha} + \frac{1}{\beta} = ?$

$\frac{\beta + \alpha}{\alpha\beta}$

$\Rightarrow \frac{3/\cancel{4}}{-7/\cancel{4}} \Rightarrow -\frac{3}{7}$

#LP: If α and β are zeroes of the polynomial $p(x) = x^2 - 2x - 1$, then find the value of $\frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta$.

(2025)

$\boxed{x^2 - 2x - 1}$ α, β

$\alpha + \beta = -\frac{(-2)}{1}$

$\boxed{\alpha + \beta = 2}$

$\alpha \cdot \beta = -\frac{1}{1}$

$\boxed{\alpha \cdot \beta = -1}$

$\Rightarrow \frac{1}{2\alpha} + \frac{1}{2\beta} + 3(\alpha\beta)$

$\Rightarrow \frac{1}{2}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3(-1)$

$\Rightarrow \frac{1}{2}\left(\frac{\beta + \alpha}{\alpha\beta}\right) - 3$

$\Rightarrow \frac{1}{\cancel{2}}\left(\frac{\cancel{2}}{-1}\right) - 3$

$\Rightarrow -1 - 3 \Rightarrow -4$

#LP: If α and β are the zeros of the polynomial $\boxed{25x^2 - 16}$, then $\boxed{\alpha^2 + \beta^2}$ is:

(A) 32 /25

(B) 25 /32

(C) 25 /16

(D) 16 /25

$\boxed{25x^2 - 16}$ α, β

$\alpha + \beta = -(0)$

$\boxed{\alpha + \beta = 0}$

$\alpha \cdot \beta = \frac{-16}{25}$

$\boxed{\alpha \cdot \beta = -\frac{16}{25}}$

$\alpha^2 + \beta^2 \Rightarrow (\alpha+\beta)^2 - 2\alpha\beta$

$\Rightarrow (0)^2 - 2\left(\frac{-16}{25}\right)$

$= \frac{32}{25}$

$(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$

$\boxed{(\alpha+\beta)^2 - 2\alpha\beta = \alpha^2 + \beta^2}$

#LP: If α, β are the zeroes of the quadratic polynomial $f(x) = 4x^2 - 5x - 1$. , find the value of $\alpha^2\beta + \alpha\beta^2$.

$\boxed{4x^2 - 5x - 1}$ α, β

$\alpha + \beta = -\frac{(-5)}{4}$

$\alpha + \beta = \frac{5}{4}$

$\alpha \cdot \beta = \frac{-1}{4}$

$\alpha^2\beta + \alpha\beta^2$

$\Rightarrow \alpha\beta(\alpha + \beta)$

$\Rightarrow -\frac{1}{4} \times \frac{5}{4}$

$\Rightarrow -\frac{5}{16}$

#LP: . If 'α' and 'β' are the zeroes of the polynomial $p(y) = y^2 - 5y + 3$, then find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$. (2025)

$$\boxed{y^2 - 5y + 3} \; \alpha, \beta$$

$$\alpha + \beta = \frac{-(-5)}{1} \qquad \alpha \cdot \beta = \frac{3}{1}$$

$$\boxed{\alpha + \beta = 5} \qquad \boxed{\alpha \cdot \beta = 3}$$

(ii) $\alpha^3 + \beta^3 = ??$

$$\alpha^4\beta^3 + \alpha^3\beta^4$$

$$\Rightarrow \alpha^3\beta^3(\alpha + \beta)$$

$$\Rightarrow (\alpha\beta)^3 (5)$$

$$\Rightarrow (3)^3 (5)$$

$$\Rightarrow 27 \times 5 \Rightarrow 135$$

(ii) $\alpha^3 + \beta^3$

$\Rightarrow (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$

$\Rightarrow (5)^3 - 3(3)(5)$

$\Rightarrow 125 - 45$

$\Rightarrow 80$

$(\alpha+\beta)^3 = \underbrace{\alpha^3+\beta^3} + 3\alpha\beta(\alpha+\beta)$

$$\boxed{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = \alpha^3+\beta^3}$$

$\alpha^{-1} + \beta^{-1}$

$\frac{1}{\alpha} + \frac{1}{\beta}$

#LP: Find the value of k such that the polynomial $\boxed{x^2 - (k+6)x + 2(2k-1)}$ has sum of its zeroes equal to half of their product.

ATQ

$$SOR = \frac{1}{2} POR$$

$$\frac{-b}{\cancel{a}} = \frac{1}{2}\left(\frac{c}{\cancel{a}}\right)$$

$$-b = \frac{1}{2} \times c$$

$$-(-(k+6)) = \frac{1}{\cancel{2}} \times \cancel{2}(2k-1)$$

$$k+6 = 2k-1$$

$$6+1 = 2k-k \Rightarrow k=7$$

#LP: If one zero of the polynomial $p(x) = 6x^2 + 37x - (k - 2)$ is reciprocal of the other, then find the value of k. (CBSE 2023)

$$\boxed{6x^2 + 37x - (k-2)} \quad \alpha, \ \frac{1}{\alpha}$$

$$SOR = -\frac{b}{a}$$

$$\alpha + \frac{1}{\alpha} = \frac{-(37)}{6}$$

$$POR = \frac{c}{a}$$

$$\cancel{\alpha} \cdot \frac{1}{\cancel{\alpha}} = \frac{-(k-2)}{6}$$

$$6 = -k + 2$$

$$6 - 2 = -k$$

$$k = -4$$

#LP: Find the value of p for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other.

$$\boxed{px^2 - 14x + 8 = 0} \quad \alpha, \ 6\alpha$$

$$SOR = -\frac{b}{a}$$

$$\alpha + 6\alpha = -\frac{(-14)}{p}$$

$$\cancel{7}\alpha = \frac{\cancel{14}^{\,2}}{p}$$

$$\boxed{\alpha = \frac{2}{p}} \quad 1$$

$$POR = \frac{c}{a}$$

$$(\alpha)(6\alpha) = \frac{8}{p}$$

$$6\alpha^2 = \frac{8}{p}$$

$$6\left(\frac{2}{p}\right)^2 = \frac{8}{p}$$

$$\cancel{6}^{\,3} \times \frac{\cancel{4}}{p^{\cancel{2}}} = \frac{\cancel{8}^{\,2}}{\cancel{p}}$$

$$\boxed{p = 3}$$

#LP: If one zero of the quadratic polynomial $f(x) = \boxed{4x^2 - 8kx - 9}$ is negative of the other, find the value of k.

$$\boxed{4x^2 - 8kx - 9} \begin{cases} \alpha \\ -\alpha \end{cases}$$

SOR $= -\frac{b}{a}$

$(\alpha) + (-\alpha) = \frac{-(-8k)}{4}$

$0 = \frac{\cancel{8}^{2}k}{\cancel{4}}$

$\boxed{k=0}$

POR $= \frac{c}{a}$

#LP: If the zeroes of the polynomial $x^2 + px + q$ are double in value to the zeroes of $2x^2 - 5x - 3$, find the value of p and q. (2012)

$$\boxed{x^2 + px + q} \quad 2\alpha,\ 2\beta$$

$$SOR = -\frac{b}{a}$$

$$2\alpha + 2\beta = -\frac{p}{1}$$

$$2(\alpha + \beta) = -p$$

$$\cancel{2} \times \frac{5}{\cancel{2}} = -p$$

$$\boxed{p = -5}$$

$$POR = \frac{c}{a}$$

$$2\alpha \cdot 2\beta = \frac{q}{1}$$

$$4\underbrace{\alpha\beta} = q$$

$$\cancel{4}^{2} \times -\frac{3}{\cancel{2}} = q$$

$$\boxed{q = -6}$$

$$\boxed{2x^2 - 5x - 3} \quad \alpha,\ \beta$$

$$\alpha + \beta = -\frac{(-5)}{2}$$

$$\alpha + \beta = 5/2$$

$$\alpha \cdot \beta = \frac{-3}{2}$$

Making of a Quadratic Polynomial :

$$\text{Quad} \Rightarrow \boxed{\overset{\in R}{k}\left[x^2 - Sx + P\right]}$$

S = Sum of zeroes

P = product of zeroes

#LPt Form a quad. $S = 4$, $P = -2$

$\rightarrow k(x^2 - Sx + P)$

$\Rightarrow k(x^2 - 4x + (-2))$

$\Rightarrow \boxed{k(x^2 - 4x - 2)}$

$k = 1 \Rightarrow \boxed{x^2 - 4x - 2}$

$k = 2 \rightarrow 2x^2 - 8x - 4$

$k = 3 \Rightarrow 3x^2 - 12x - 6$

$\vdots \qquad \vdots$

There exist infinite Quadratic Polynomials having two zeroes.

#LP: Find a quadratic polynomial, the sum and product of whose zeroes are 0 and $-\sqrt{2}$ respectively. (2015)

$$\left.\begin{aligned} S &= 0 \\ P &= -\sqrt{2} \end{aligned}\right\} \longrightarrow k(x^2 - Sx + P)$$

$$k(x^2 - 0x + (-\sqrt{2}))$$

$$k(x^2 - \sqrt{2})$$

$$k = 1 \longrightarrow \boxed{x^2 - \sqrt{2}}$$

#LP: Form a quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. (2012)

$\alpha = 3 + \sqrt{2}$

$\beta = 3 - \sqrt{2}$

$S = (3 + \cancel{\sqrt{2}}) + (3 - \cancel{\sqrt{2}})$

$\boxed{S = 6}$

$P = (3 + \sqrt{2})(3 - \sqrt{2})$

$P = (3)^2 - (\sqrt{2})^2$

$P = 9 - 2 \Rightarrow 7 = P$

$Q = k[x^2 - Sx + P]$

$Q = k[x^2 - 6x + 7]$

let $k = 1 \Rightarrow (x^2 - 6x + 7)$

#1Q: If α, β are roots of $\boxed{x^2+5x+5=0}$, then find a quadratic whose roots are $1+\alpha$, $1+\beta$.

$$\boxed{x^2+5x+5=0} \begin{cases} \alpha \\ \beta \end{cases}$$

$\alpha+\beta = \dfrac{-(5)}{1}$ | $\alpha\cdot\beta = \dfrac{5}{1}$

$\boxed{\alpha+\beta=-5}$ | $\boxed{\alpha\cdot\beta=5}$

$$Q = k\left(x^2 - Sx + P\right)$$

$$S = (1+\alpha)+(1+\beta)$$
$$= 2+\alpha+\beta \Rightarrow 2+(-5) = -3 = S$$

$$P = (1+\alpha)(1+\beta)$$
$$= 1+\underbrace{\beta+\alpha}+\alpha\beta$$
$$= 1+(\cancel{-5})+\cancel{5} = 1 = P$$

$\therefore Q \rightarrow k[x^2 - Sx + P]$

$= 1(x^2 - (-3)x + 1]$

$Q = \boxed{x^2 + 3x + 1}$

#LP:

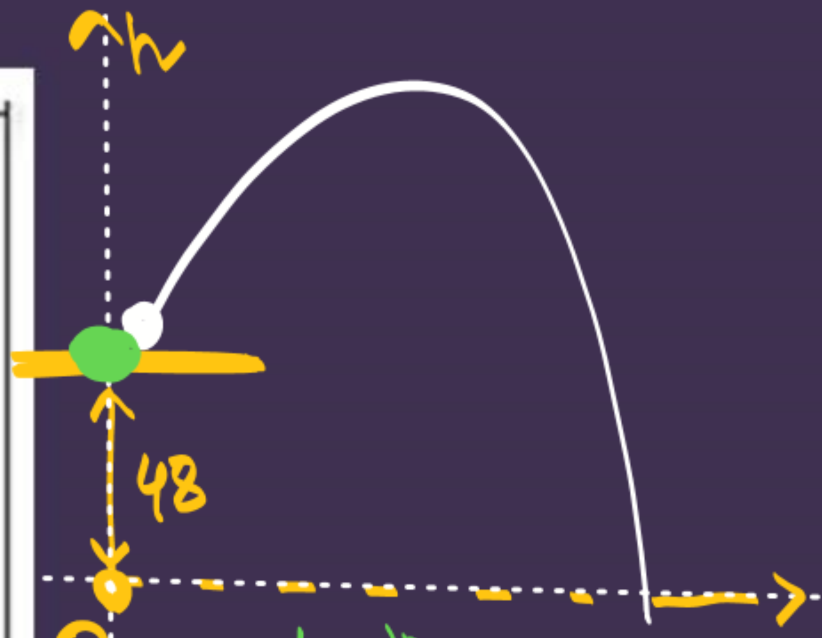
Q41-Q45 are based on Case Study -1

Case Study -1

The figure given alongside shows the path of a diver, when she takes a jump from the diving board. Clearly it is a parabola.

Annie was standing on a diving board, 48 feet above the water level. She took a dive into the pool. Her height (in feet) above the water level at any time 't' in seconds is given by the polynomial h(t) such that

$$\boxed{h(t) = -16t^2 + 8t + k.}$$



$(0, 48)$

$t = 0$

$h = 48$

$48 = -16(0)^2 + 8(0) + k$

$\boxed{48 = k}$

What is the value of k? — 1

(a) 0
(b) - 48
(c) 48
(d) 48/-16

$\therefore \boxed{h = -16t^2 + 8t + 48}$

At what time will she touch the water in the pool? — 1

$t = ?$

(a) 30 seconds
(b) 2 seconds
(c) 1.5 seconds
(d) 0.5 seconds

$h = 0$

$0 = -16t^2 + 8t + 48$

$0 = -8(2t^2 - t - 6)$

$\boxed{2t^2 - t - 6 = 0}$

$2t^2 - 4t + 3t - 6 = 0$

$t = 2, \cancel{-3}$

Rita's height (in feet) above the water level is given by another polynomial $p(t)$ with zeroes -1 and 2. Then $p(t)$ is given by- (a) $t^2 + t - 2$. (b) $t^2 + 2t - 1$ (c) $24t^2 - 24t + 48$. (d) $-24t^2 + 24t + 48$.	1
A polynomial $q(t)$ with sum of zeroes as 1 and the product as -6 is modelling Anu's height in feet above the water at any time t(in seconds). Then $q(t)$ is given by (a) $t^2 + t + 6$ (b) $t^2 + t - 6$ (c) $-8t^2 + 8t + 48$ (d) $8t^2 - 8t + 48$ $\left.\begin{array}{l} S = 1 \\ P = -6 \end{array}\right\} \rightarrow k(x^2 - 1x - 6)$ $\boxed{k(x^2 - x - 6)}$	1
The zeroes of the polynomial $r(t) = \boxed{-12t^2 + (k-3)t + 48}$ are negative of each other. Then k is HW α $-\alpha$ (a) 3 (b) 0 (c) -1.5 (d) -3	1

#LP:

In a pool at an aquarium, a dolphin jumps out of the water travelling at 20 cm per second. Its height above water level after t seconds is given by $h = 20t - 16t^2$.

Based on the above, answer the following questions :

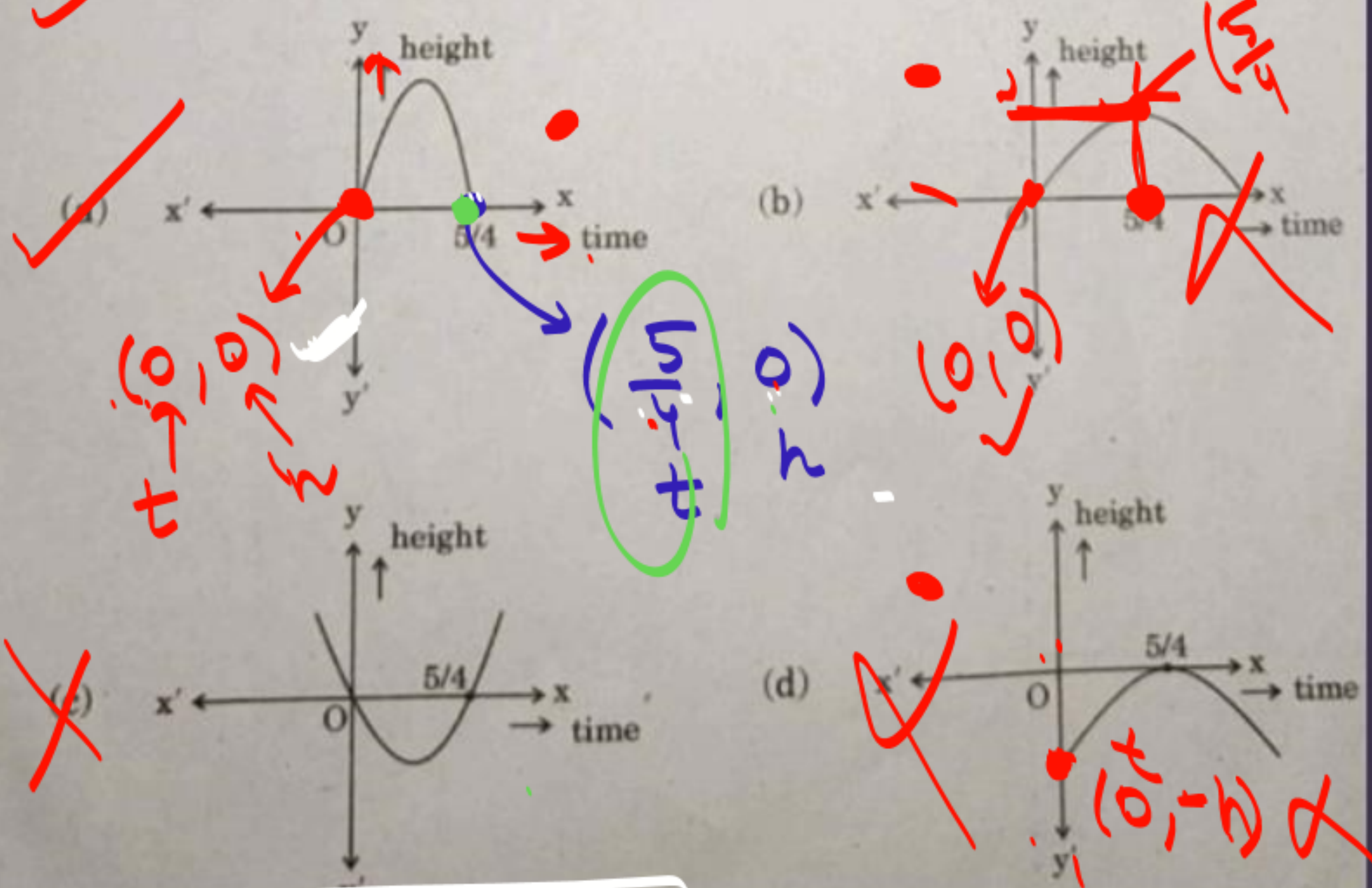
(i) Find zeroes of polynomial $p(t) = 20t - 16t^2$.

$$\boxed{h = 20t - 16t^2}$$

$t = 5/4$

$h = 0$

(ii) Which of the following types of graph represents p(t) ?



$$\boxed{p(t) = 20t - 16t^2}$$

$20t - 16t^2 = 0$

$(4t)(5 - 4t) = 0$

$4t = 0$ | $5 - 4t = 0$

$t = 0$ | $t = 5/4$

$a < 0 \rightarrow$ downwards

$$\boxed{ax^2+bx+c}$$

Upwards

$$\boxed{a>0}$$

downwards.

$$\boxed{a<0}$$

ii) What would be the value of h at $t = \frac{2}{3}$? 1

iii) How much distance has the dolphin covered before hitting the water level again? 2

$$h = 20t - 16t^2$$

$$t = 2/3 \rightarrow h = 20\left(\frac{2}{3}\right) - 16\left(\frac{2}{3}\right)^2$$

$$= \frac{40}{3} - 16 \times \frac{4}{9}$$

$$= \frac{40}{3} - \frac{64}{9} \Rightarrow \frac{120-64}{9} \Rightarrow \frac{56}{9}$$

d = ?

at $t = \frac{5}{4}$ sec

ATQ 1 sec → 20 cm

$\frac{5}{4}$ sec → $\left(\frac{5}{\cancel{4}} \times \cancel{20}\right)$ cm

→ 25 cm

10/14 completed!

Thank you Coodiess !!